RUPINDER PAL SINGH

SENIOR FRONTEND ENGINEER

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PROFESSIONAL SUMMARY

Senior Frontend Engineer with 8+ years of extensive experience building scalable, high-performing enterprise web applications using Next.js, React.js, and TypeScript. Specialized in modern frontend architecture, state management optimization, and delivering seamless, accessible user experiences. Adept at bridging complex backend microservices with intuitive interfaces, while actively integrating modern AI capabilities (OpenAI API, RAG workflows, vector embeddings) into production client applications to boost user engagement and automation.

TECHNICAL EXPERTISE

Frameworks & Libraries: Next.js, React.js, NestJS, Redux Toolkit, Tailwind CSS, HTML5, CSS3

Languages: JavaScript (ES6+), TypeScript, SQL, HTML/CSS

AI & Machine Learning: OpenAI API, RAG Pipelines, Vector Databases (Supabase), Embeddings, GitHub Copilot

Backend & APIs: Node.js, Express.js, RESTful APIs, GraphQL

Testing & Quality: Jest, Cypress, React Testing Library, Automated CI/CD Testing

Cloud & DevOps: AWS, Azure, Git, GitHub, GitLab, Bitbucket, Docker

PROFESSIONAL EXPERIENCE

Senior Front End Engineer | LGS-IBM-BELL

Canada | April 2025 – Present

- Architected and delivered high-impact customer-facing features for Bell's e-commerce platform using Next.js and React.js, elevating responsiveness and performance by 25%.
- Spearheaded component modularization and lazy-loading strategies, significantly optimizing Core Web Vitals metrics (LCP, FID, CLS) for enhanced SEO and user retention.
- Integrated state-of-the-art AI search and recommendation features leveraging OpenAI APIs and vector-based retrieval systems, transforming customer product discovery.
- Mentored junior and intermediate engineers, establishing frontend code standards, rigorous code review protocols, and automated accessibility (a11y) guidelines.

Front End Engineer | LGS-IBM-BELL

Canada | January 2022 – March 2025

- Designed and implemented key e-commerce features for Bell using React.js and TypeScript, cutting component re-render overhead and boosting overall page responsiveness by 20%.
- Refactored complex legacy codebase into Redux Toolkit architecture, reducing redundancy, simplifying global state workflows, and improving maintainability across teams.
- Streamlined integration of robust RESTful APIs, optimizing product filtering, real-time cart management, and secure checkout processing pipelines.
- Collaborated closely with QA engineering to implement comprehensive end-to-end testing suites via Cypress and Jest, dropping production bug rates by 30%.

PROFESSIONAL EXPERIENCE (CONTINUED)

Front End Developer | ENRAGE TECH (Multiple Clients) India | June 2015 – September 2018

- Engineered high-performance, fully responsive web applications using modern HTML5, CSS3, and JavaScript, ensuring flawless cross-browser compatibility.
- Developed extensive libraries of reusable React.js components to accelerate delivery cycles and enforce design system consistency across diverse client projects.
- Partnered with backend engineers to integrate scalable RESTful APIs, enabling seamless real-time data binding and dynamic client-side rendering.
- Profiled and refactored inefficient frontend scripts and legacy stylesheets, reducing average page load speeds by 25% across multiple enterprise portals.

EDUCATION & CERTIFICATIONS

AEC IT Programmer Analyst
College LaSalle — Montreal, QC
#Full_Stack_Development #Node.js #React #MongoDB #Git #Advanced_Algorithms

Bachelor of Science in Computer Science
Guru Nanak Dev University — Amritsar, Punjab
#Object_Oriented_Programming #Data_Structures #Database_Management #Software_Engineering

KEY PROJECTS & ARCHITECTURE HIGHLIGHTS

Enterprise AI RAG Assistant Integration | Personal / Client Showcase Next.js, TypeScript, OpenAI, Supabase Vector

- Engineered a high-throughput retrieval-augmented generation (RAG) assistant interface utilizing Next.js App Router and Server Actions.
- Implemented custom vector embedding searches via Supabase to deliver context-aware, low-latency technical support responses for enterprise users.

E-Commerce Optimization Toolkit | Open Source Contributions React.js, Tailwind CSS, Redux Toolkit

- Created modular UI component kits emphasizing accessibility (WCAG 2.1 AA compliance) and lightning-fast state synchronization for high-traffic retail portals.